Change Detection of Open Pit Mining Areas in Remote Sensing Images Based on Coupled Local-Global Features

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2024 5th International Conference on Machine Learning and Computer Application (ICMLCA) | 979-8-3315-3033-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICMLCA63499.2024.10754176

***Abstract*—In this paper, a coupled local-global feature-based change detection method Swin-LGCNet is proposed for remote sensing image open pit mining areas The model interactively fuses local and global features extracted by ResNet and Swin Transformer through the local-global feature coupling module, which effectively improves the feature expression capability. The Mish activation function is used instead of ReLU, which reduces the information loss and enhances the model's ability to capture complex changing features. Meanwhile, the differential operation is introduced to further enhance the difference between the dual-time phase features and significantly reduce the leakage detection rate. The experimental results show that Swin-LGCNet outperforms the traditional network in both global information extraction and local detail performance, and the detection accuracy is significantly improved compared with the comparison networks, such as PSPNet, Deeplabv3+, UNet and UNet++, especially in boundary identification and noise suppression. The method makes full use of the temporal sequence of remote sensing images, and provides efficient technical support for dynamic change monitoring and ecological protection in open pit mining areas.**

Keywords-local-global features, remote sensing imagery, open pit mining area, change detection.

I. INTRODUCTION

Mineral resources are an important support for China's economic development, but the ecological problems in mining areas are becoming more and more serious due to the long-term development in a more rough way, coupled with the fragility of the ecological environment in mining areas. This not only threatens the life and property safety of local residents, but also hinders the sustainable development of mines. Remote sensing change detection extracts surface type change information by processing and comparing remote sensing images of different periods in the same area. This technique can grasp the information of surface cover and land use change in time, which is of great practical significance in land planning, ecological protection and other practical applications [1].

With the rapid development of deep learning technology, some scholars have begun to utilize its advantages to carry out research on change detection in mining areas. For example, Li Jun et al. Realized the comparative analysis of vegetation damage recognition in Shendong coal base based on the improvement based on LandTrendr and CCDC algorithms [2]. Guo Mengli et al. Constructed densely connected ASSP modules with different null rates based on the original DeepLabV3+ model to optimize the effect of extracting the road network from open pit mines [3]. Bi Weihua et al. Constructed an application platform for mine surface change detection based on the ResNet twin network combined with the Cesium open source 3D engine framework, but the platform still needs to rely on human-computer interaction with a limited degree of automation [4]. Ruan Thuan-Liang et al. then improved the DeeplabV3+ model based on DeeplabV3+ model to complete the detection of vegetation cover changes in open pit mining areas [5]. However, most of the above methods are based on CNN architecture, and their extracted features are mainly focused on local information, which makes it difficult to comprehensively measure the importance of global features [6]. In contrast, the Transformer architecture, with its powerful global context modeling capability [7-8], is better able to characterize the changes in the region of interest, but may have limitations in locating the boundaries of the change region due to its lack of accurate capturing of details.

Inspired by previous studies, this paper proposes a remote sensing image open pit change detection method (Swin-LGCNet) based on coupling of local and global features. In Swin-LGCNet, the local-global feature coupling module (LGCM) enhances the local detail information and global context information respectively through different attention mechanisms, and then the features are interactively fused to improve the feature expression ability. In addition, to address the problem that the use of ReLU activation function for ResNet may lead to partial information loss during the training process, this paper introduces the Mish activation function instead of ReLU to reduce the information loss and better extract the complex change features of the mining area, so as to improve the generalization ability of the model. Through the training

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and accuracy verification of the dataset, the method significantly improves the accuracy of change detection in mining areas while simplifying the manual design of the feature extraction process.

II. Network Infrastructure

A. General overview of the network

Fig. 1 shows the overall architecture of Swin-LGCNet. The network adopts an encoder-decoder structure, and in the encoder part, a parallel ResNet branch and a Swin Transformer branch are set up, which are used to extract multi-scale local features and global features of dual-phase remote sensing images, respectively. At each scale, the LGCM module couples the local and global features to generate multi-scale features with both local details and global spatial information. At the last layer of coding, the multiscale features are further enhanced using the Hollow Space Pyramid (HSP) [9]. In the decoder part, the four feature maps generated by the encoder are summed element-by-element to obtain the multi-scale variation map. In order to cope with the complex background in the remote sensing image, a multilateral output fusion strategy is adopted. Then the difference operation is performed on the change maps at each scale to generate differential change maps with stronger discriminative ability to reduce the background noise interference, and finally the final change detection results are output by fusing the differential change maps at all scales and passing through the classifier.

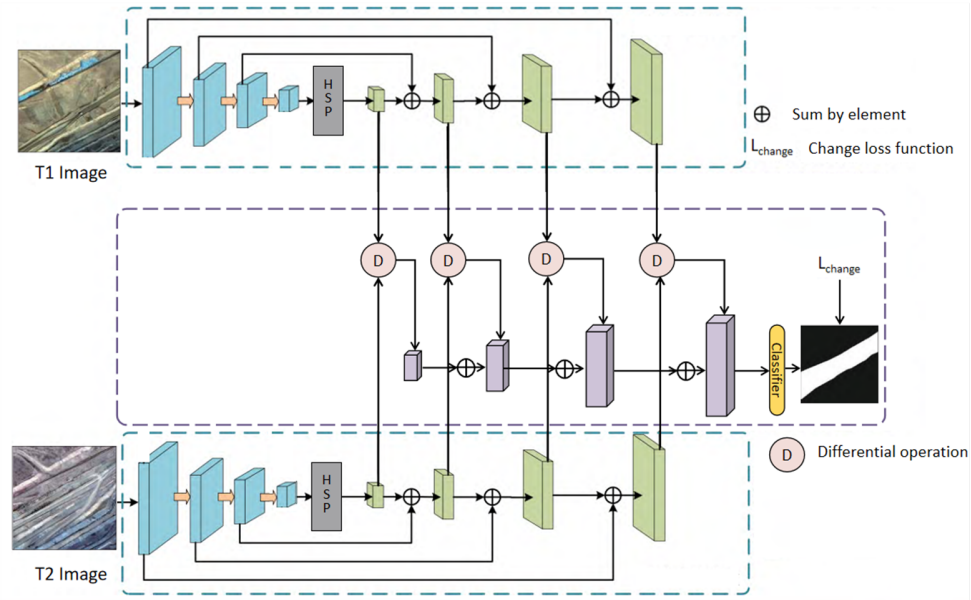


Fig. 1 General architecture of Swin-LGCNet model

B. Visual Swin Transformer Structure

Swin Transformer is an efficient visual transformer designed for visual tasks, which solves the efficiency problem of traditional transformers in computing large-scale images by introducing the attention mechanism of sliding window.The overall architecture of Swin Transformer is based on a hierarchical design, which progressively extracts features from smaller localized regions, and then gradually expands the receptive field through the The overall architecture of Swin Transformer is based on a hierarchical design, where features are gradually extracted from smaller local areas, and then the sensory field is gradually expanded through hierarchical feature fusion. In each layer, the image is divided into fixed-size non-overlapping windows, and the image blocks within the windows are computed by the local attention mechanism [10]. Different from the traditional Transformer, Swin Transformer introduces the mechanism of sliding windows between layers, which makes the window positions between different layers offset and enhances the information exchange between different windows. In addition, with the depth of the network, the size of the window and the resolution of the feature map will gradually change, realizing multi-scale feature extraction from local to global. Thanks to this hierarchical design, Swin Transformer significantly reduces the computational overhead while guaranteeing the model expression capability.

C. Local Global Feature Coupling Module (LGCM)

In order to effectively integrate the coded features extracted from the ResNet branch and the Swin Transformer branch, this paper proposes a local-global feature coupling module (LGCM) based on the attention mechanism. As shown in Fig. 2, the structure of LGCM consists of several branches: the local features extracted from the ResNet branch use the spatial attention mechanism in CBAM to enhance the details and suppress irrelevant variations and noise interference; while the global features extracted from the Swin Transformer branch use the channel attention mechanism to further enhance the global information. Fig. 2 shows the process of local and global features being processed by spatial attention and channel attention, respectively, with an additional branch to realize the fusion of local and global features by element-by-element summation.

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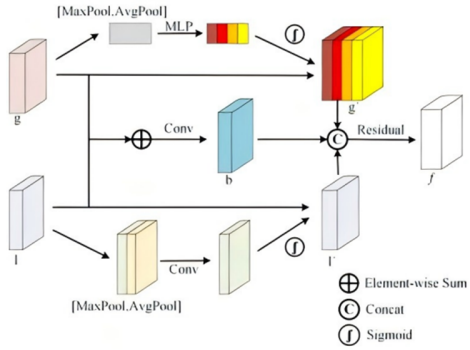


Fig. 2 Structure diagram of local-global feature coupling module

D. *Mish activation function*

Mish is a smooth, continuous and non-monotonic activation function with a curve similar to ReLU but smoother, thus facilitating deeper transfer of information to the neural network. Unlike ReLU, Mish does not set the gradient to zero when the input is negative, allowing small negative gradients to flow and reducing neuron inactivation, which improves model training, accuracy, and generalization. The Mish function is bounded in the negative direction and unbounded in the positive direction, and takes values in the range $[\approx -0.31, +\infty)$. Its mathematical expression is:

$$f(x) = x \cdot \tanh[\ln(1+e^{x})] \tag{1}$$

Where f (x) is the Mish activation function, x is the independent variable, tanh denotes the hyperbolic tangent function, and ln is the natural logarithm.

III. Experimental Data

To verify the effectiveness of the method in this paper, two Gaofen-2 (GF2) remote sensing images from April 2020 and July 2023 of an open-pit mine in Inner Mongolia were selected. After cropping, the image area size of the mine is 4000×4000 pixels. The data preprocessing includes radiometric calibration, atmospheric correction and ortho rectification, and finally the 1 m spatial resolution panchromatic image is fused with the 4 m multispectral image to produce a 1 m resolution fusion image containing red, green and blue bands. The mine has been in operation since 2001, and the reclamation of the discharge site will be basically completed by 2022, with significant changes in the surface cover type compared to the pre-reclamation period.

IV. Experiments and Results

This experiment was done on a desktop computer with Intel Core i9-10900K processor, 32G RAM and NVIDIA GeForce RTX3090 graphics card. The deep learning framework is Paddle, the programming language is Python, and the development environment is Visual Studio Code.

A. *Parameterization*

In the model training process, Binary Crossentropy (Binary Crossentropy) is used as the loss function, and IoU (Intersection over Union) is used as the evaluation index. The optimizer selects Adam, the initial learning rate is set to 0.001, and the maximum number of training rounds is 200. when the loss in the validation set no longer decreases, the learning rate is adjusted according to the decay factor 0.2, i.e., the new learning rate is the initial value multiplied by the decay factor. The tolerance of Early Stopping strategy (Early Stopping) is set to 10 epochs, i.e., when the validation set loss does not show a decrease in 10 consecutive epochs, the training is terminated to avoid overfitting.

B. *Experimental results*

In order to verify the effectiveness of the method in this paper, a series of comparison experiments are designed to detect the open pit using semantic segmentation and change detection networks such as PSPNet, Deeplabv3+, UNet and UNet++, respectively, and the results are compared with Swin-LGCNet. The parameter configurations of each comparison network refer to the relevant literature. Table 1 summarizes the accuracy performance of each network in the mine change detection task. The commonly used accuracy evaluation metrics, including Overall Accuracy (OA), Kappa coefficient, Omission (Omission) and False Detection (Commission), were calculated by confusion matrix.

Table 1 Accuracy of Each Network

Evaluating Indicator	Kappa Coefficient	OA	Omission	Commission
Swin-LGCNet	0.8237	0.9681	0.0081	0.2085
PSPNet	0.6905	0.9254	0.1782	0.3596
Deeplabv3+	0.7299	0.9306	0.1564	0.2971
UNet	0.7986	0.9567	0.0897	0.2527
UNet++	0.8051	0.9491	0.1695	0.2151

Swin-LGCNet is applied to mine change detection, and its Kappa, OA, Omission, and Commission metrics all achieve significant improvements over the four comparison networks. Specifically, Kappa improves by 1.86% to 13.32%, OA improves by 1.9% to 4.27%, Omission reduces by 7.78% to 17.01%, and Commission reduces by 0.66% to 15.11%. In order to show the detection effect more intuitively, a piece of image block was selected for visualization, and the results are shown in Figure 3. Compared with the detection results of PSPNet and Deeplabv3+, Swin-LGCNet shows better overall performance. Although PSPNet and Deeplabv3+ are able to extract global information, they still lack local details. Compared with UNet, Swin-LGCNet effectively avoids the rough edges of image blocks and the black edges produced by splicing. And compared with UNet++, Swin-LGCNet shows a smoother and more complete mine boundary structure with no obvious splicing problem.

In summary, Swin-LGCNet combines the advantages of CNN and Swin Transformer, which not only performs well in global information extraction, but also outperforms other methods in local details and mine boundary detection, and achieves more accurate change detection results.

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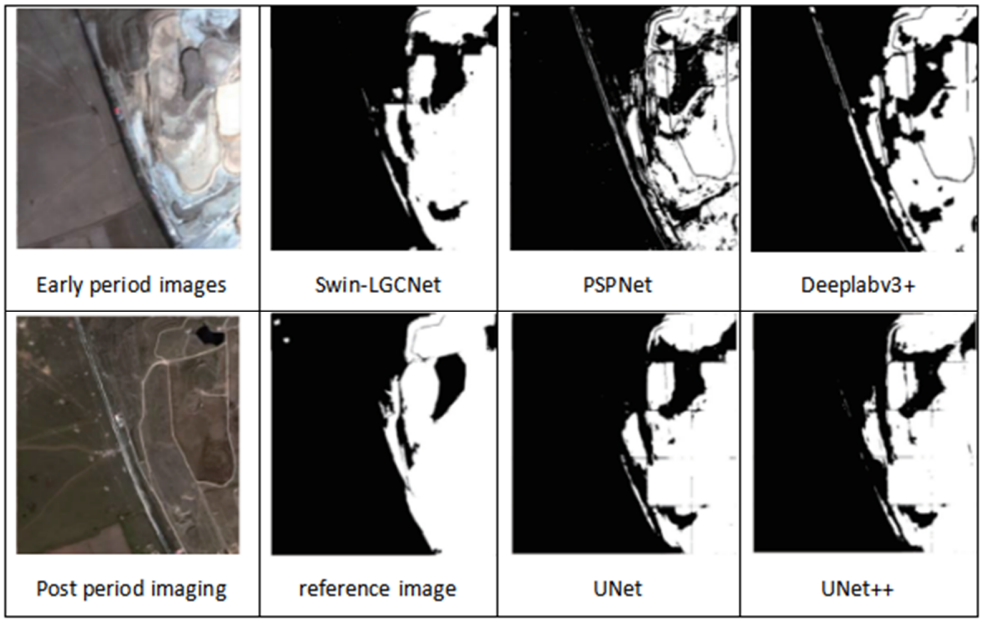


Fig. 3 Change detection results

V. Conclusions

Aiming at the change detection task of open pit mining areas in high-resolution remote sensing images, this paper designs a Swin-LGCNet network model based on local-global feature coupling. The model improves the feature expression capability by interactively fusing the parallel local and global features through the local-global feature coupling module; the introduction of the Mish activation function reduces the information loss and better extracts the complex change features of the mining area; and the difference operation effectively enhances the difference between the bi-phasic features and significantly reduces the leakage detection rate, which demonstrates the superior performance of Swin-LGCNet. The model has excellent performance in both global information capture and local detail performance.

The method in this paper is applicable to the change detection of remote sensing images of open-pit mining areas, which makes full use of the temporal characteristics of remote sensing images to dynamically extract the information of land use change in mining areas, and helps to promote the ecological protection and restoration. Compared with the traditional feature extraction methods, the network model combining CNN and Transformer can extract remote sensing image features more efficiently, provide technical support for image interpretation, and significantly improve the interpretation efficiency.

Acknowledgements

This research was supported by The Fundamental Research Funds for Beijing University of Civil Engineering and Architecture (No.ZF16064).

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